Installing RSAS 4.0

This guide covers every supported way to install **RSAS (RNA Structure Analysis Suite) 4.0** on Linux, macOS and Windows: a Conda environment, a Docker image, and native Python installations for each operating system. It also covers the optional RNArobo and Knotty engines, verification, and troubleshooting.

Versão em português: INSTALAR.md.

> **Tip**
>
> **Short version.** Install Miniforge, then run `conda env create -f environment.yml` (Linux/macOS) or `conda env create -f environment-windows.yml` (Windows), `conda activate rsas` and `python main.py`.

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1. What gets installed

Component	Purpose	Source
Python 3.10 – 3.13	Runtime (3.11 recommended)	Conda, python.org, or the OS package manager
ViennaRNA ≥ 2.6	RNA folding: MFE, partition function, base-pair probabilities (Python module `RNA`)	Bioconda, or the official PyPI wheel
customtkinter 5.x + Tk 8.6	Desktop interface	PyPI / Bioconda; Tk ships with Python
numpy, openpyxl, biopython	Core analysis and Excel output	PyPI / conda-forge
python-pptx, matplotlib, pillow, lxml	RNAfold Deck: structure drawings and PowerPoint	PyPI / conda-forge
RNArobo 2.1.0 *(optional)*	RNArobo Search page (structural motif search)	Bundled for macOS and Windows; Linux executable from the authors; source (§8)
Knotty *(optional)*	Pseudoknot Finder page	Bundled for macOS and Windows; build from source on Linux (§8)

RSAS needs **no internet connection and no web browser** at run time. All folding, including the RNAfold Deck structure drawings, is done locally with ViennaRNA. The only exception is the Sequence Extractor's optional *Fetch from NCBI* tab.

2. Choosing an installation method

Method	Best for	Desktop GUI	Command line	RNArobo / Knotty
A. Conda	Most users, on any OS. One command, reproducible	✅	✅	macOS, Windows: bundled. Linux: download/build
B. Docker	Servers, HPC front-ends, pipelines, reproducible batch runs	⚠️ needs an X server	✅	Mount Linux builds into the container
C. Native (venv + pip)	Users without Conda, or who manage Python themselves	✅	✅	Same as Conda

> **Note**
> All three methods install the same code and the same ViennaRNA version (2.7.2 at the time of writing). Results are identical across methods and operating systems.

3. System requirements

	Minimum	Recommended
Operating system	Windows 10 (64-bit), macOS 11 Big Sur, or a Linux distribution with glibc ≥ 2.24 (Ubuntu 18.04+, Debian 10+, RHEL/Rocky 8+)	Windows 11, macOS 13+, Ubuntu 22.04/24.04
CPU architecture	x86_64, or ARM64 on macOS (Apple Silicon) and Linux	Any of these
Python	3.10	3.11 or 3.12
Memory	4 GB	8 GB or more for large batches and partition-function runs
Disk	1.5 GB (Conda environment)	3 GB
Display	Needed only for the GUI	—

> **Warning**
> Windows on ARM is not supported natively because there is no ViennaRNA build for it. Use WSL2 or Docker instead.

4. Getting the source code

With Git (recommended, makes updates easy):

```
git clone https://github.com/<owner>/RSAS.git
cd RSAS
git checkout v4.0.0
```

Replace `<owner>` with the GitHub account that hosts your copy of RSAS.

Without Git: on the repository page, click **Code → Download ZIP**, extract the archive, and open a terminal in the extracted folder.

All commands below are run **from the repository root**, the folder that contains `main.py`.

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5. Method A: Conda (recommended)

Conda installs Python, Tk and ViennaRNA as pre-built binaries, so no compiler and no system packages are needed. The repository provides two environment files:

File	Platforms	ViennaRNA comes from
`environment.yml`	Linux (x86_64), macOS Intel, macOS Apple Silicon	Bioconda (native build)
`environment-windows.yml`	Windows 10/11 (x86_64)	Official PyPI wheel (Bioconda has no Windows build)

5.1 Install a Conda distribution

Install **Miniforge**. It is a minimal installer that uses the community `conda-forge` channel by default and includes `mamba`, a faster drop-in replacement for `conda`. Miniconda and Anaconda also work.

▶ **Linux / macOS**
▶ **Windows**

5.2 Create the environment

Linux and macOS:

```
conda env create -f environment.yml
conda activate rsas
```

Windows (Miniforge Prompt):

```
conda env create -f environment-windows.yml
conda activate rsas
```

> **Tip**
> Replace `conda` with `mamba` in the `env create` command for a faster solve.

The environment is named `rsas`. Activate it in every new terminal before running RSAS.

5.3 Install the RSAS commands (optional)

```
python -m pip install --no-deps -e .
```

This adds two commands to the environment: `rsas` (desktop application) and `rsas-cli` (command line). Without this step, use `python main.py` and `python -m RnaThermofinder` instead; they are equivalent.

5.4 Verify and run

```
python -m RnaThermofinder version     # RSAS 4.0.0 | ViennaRNA 2.7.2 | Python 3.11.x
python -m pytest -q tests             # all tests should pass
python main.py                        # start the desktop application
```

Continue with §8 if you need RNArobo Search or the Pseudoknot Finder.

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6. Method B: Docker

The Docker image contains the complete Linux environment: ViennaRNA, the RSAS command line and the RNAfold Deck. The image is built from the same `environment.yml` used by Method A, and the build runs the test suite, so a successful build is a verified installation.

6.1 Prerequisites

- **Linux:** Docker Engine 20.10 or later (installation instructions). Add your user to the `docker` group, or prefix the commands below with `sudo`.
- **macOS / Windows:** Docker Desktop. On Windows, enable the WSL2 back end, which is the default.

6.2 Build the image

From the repository root:

```
docker build -t rsas:4.0.0 .
```

The build takes several minutes: it downloads the Conda packages and runs the test suite. Every dependency is available for both `linux/amd64` and `linux/arm64`, so Docker builds a native image on Intel/AMD machines and on Apple Silicon alike.

6.3 Run analyses (no display needed)

The container's entry point is the RSAS command line (`python -m RnaThermofinder`). Mount the folder that holds your input files as `/data`, and write results back to it:

Linux / macOS:

```
docker run --rm -v "$PWD":/data rsas:4.0.0 version
docker run --rm -v "$PWD":/data rsas:4.0.0 analyze /data/my_sequences.fasta --out-dir /data --
cores 4
```

Windows (PowerShell):

```
docker run --rm -v "${PWD}:/data" rsas:4.0.0 analyze /data/my_sequences.fasta --out-dir /data -
-cores 4
```

Each analysis creates a new folder, `my_sequences_<YYYYmmdd_HHMMSS>/`, next to the input file. Every file inside starts with the input's name. Build an RNAfold Deck from that run as follows (replace `<run>` with the folder name):

```
docker run --rm -v "$PWD":/data rsas:4.0.0 deck \
  --results "/data/<run>/my_sequences_rna_results.xlsx" \
  --tier "Tier 1" --fold both --temp 25 --temp 37 --temp 42 \
  --out-dir "/data/<run>" --name my_sequences_RNAfold_deck
```

> **Note**
> Files written by the container belong to the container user (UID 57439). On Linux, add `--user "$(id -u):$(id -g)" -e HOME=/tmp` to `docker run` so the results belong to you.

Analysis settings saved by the GUI (output columns, scoring profiles, temperatures) live in `~/.rsas/` inside the container. To reuse your own settings, mount your folder: `-v "$HOME/.rsas":/home/mambauser/.rsas`.

6.4 Run the desktop GUI from the container (optional)

The GUI needs an X11 display. For daily GUI use, Methods A and C are simpler. Docker is mainly intended for headless work.

- ▶ **Linux (X11 or XWayland)**
- ▶ **macOS (XQuartz)**
- ▶ **Windows 11 (WSLg)**

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7. Method C: Native installation

A native installation uses a standard Python virtual environment (`venv`) and `pip`. ViennaRNA is installed from its official PyPI wheel, available for Python 3.10–3.13 on Windows x64, macOS (Intel and Apple Silicon) and Linux (x86_64 and aarch64). No compiler is needed.

7.1 Windows

Step 1. Install Python. Download **Python 3.11 or 3.12 (64-bit)** from python.org. In the installer:

- tick **Add python.exe to PATH**;
- keep **tcl/tk and IDLE** selected (required for the GUI).

Step 2. Open PowerShell in the RSAS folder (in File Explorer, *Shift + right-click → Open in Terminal*) and create the environment:

```
py -3.11 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

> **Note**
> If PowerShell refuses to run `Activate.ps1`, allow local scripts once with `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`, or use Command Prompt and run `.venv\Scripts\activate.bat`.

Step 3. Verify and run:

```
python -m RnaThermofinder version
python main.py
```

RNArobo and Knotty on Windows. Windows builds of both engines are bundled in `bin\windows\` (`rnarobo.exe`, `knotty.exe` and Knotty's `simfold\` parameters), so every page works natively; see §8.3.

Building the Windows executable

These steps produce `dist\RSAS.exe`, a single file that runs on 64-bit Windows (Intel/AMD) without Python. It already contains Python, ViennaRNA, `rnarobo.exe`, `knotty.exe` and Knotty's `simfold\` parameters. Build it on a Windows PC: PyInstaller cannot create a Windows executable on macOS or Linux, and Windows on ARM is not supported (no ViennaRNA build).

1. Install the tools (once).

- **Miniforge:** download *Miniforge3-Windows-x86_64.exe* from https://conda-forge.org/download/ and install it with the default options.
- **Git:** https://git-scm.com/download/win.

2. Open the *Miniforge Prompt* (Start menu) **and download the project:**

```
cd %USERPROFILE%
git clone https://github.com/RNAThermometer/RSAS.git
cd RSAS
```

3. Create the environment:

```
conda env create -f environment-windows.yml
conda activate rsas
python -c "import PyInstaller, RNA, tkinter, customtkinter, pptx; print('ok')"
python -m pytest -q tests
```

4. Build the executable (this takes a few minutes):

```
python build_app.py
```

The output should include `Bundling rnarobo binary: bin\windows\rnarobo.exe`, `Bundling knotty binary: bin\windows\knotty.exe`, `Bundling knotty params: ...` and `Build complete!`.

5. Test it:

```
dist\RSAS.exe
```

Check that the About page shows version 4.0.0, that **Load Example → Run** produces results, that the RNAfold Deck creates a PowerPoint file, and that the RNArobo Search and Pseudoknot Finder pages show a green banner. The first start takes a few seconds, because the single-file executable unpacks itself before it starts.

6. Distribute it. Rename the file, for example to `RSAS_4.0.0_Windows.exe`, and attach it to the GitHub release (**Releases → Draft a new release** or **Edit**). Do not commit `dist\`; it is excluded by `.gitignore`.

> **Note**
> The executable is not digitally signed, so Windows SmartScreen shows *"Windows protected your PC"*; users click **More info → Run anyway**. Antivirus programs occasionally flag PyInstaller executables by mistake; such false positives can be reported at https://www.microsoft.com/wdsi/filesubmission.

7.2 macOS

Step 1. Install Python 3.11 or 3.12 from python.org. The *universal2* installer runs natively on both Intel and Apple Silicon and includes a working Tk. After installing, double-click **Install Certificates.command** in `/Applications/Python 3.x/`.

> **Warning**
> Avoid Apple's `/usr/bin/python3` (Command Line Tools). Its Tk is outdated and the GUI renders incorrectly. If you prefer Homebrew Python, also install its Tk bindings, for example `brew install python@3.12 python-tk@3.12`.

Step 2. Create the environment (Terminal, in the RSAS folder):

```
python3.11 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

Step 3. Verify and run:

```
python -m RnaThermofinder version
python main.py
```

RNArobo and Knotty are bundled for macOS; see §8.1 for the one-time setup. If macOS reports that RSAS or one of these programs *is damaged* or *cannot be verified*, see §7.4.

Building the macOS application

These steps produce `dist/RSAS.app` and the installer `dist/RSAS_macOS.dmg`. The app contains Python, ViennaRNA, RNArobo and Knotty, and needs nothing else on the target Mac. Build it on a Mac: PyInstaller cannot create a macOS application on Windows or Linux. The app runs on Macs of the same processor type as the Mac that built it (Apple Silicon `arm64` or Intel `x86_64`).

1. Install the tools (once).

- **Miniforge:** download the installer for your Mac (Apple Silicon or Intel) from https://conda-forge.org/download/ and run it, for example `bash Miniforge3-MacOSX-arm64.sh`.
- **Git:** run `xcode-select --install` in Terminal (Apple Command Line Tools).

2. Open Terminal and download the project:

```
cd ~
git clone https://github.com/RNAThermometer/RSAS.git
cd RSAS
```

3. Create the environment:

```
conda env create -f environment.yml
conda activate rsas
python -c "import PyInstaller, RNA, tkinter, customtkinter, pptx; print('ok')"
python -m pytest -q tests
uname -m
```

`uname -m` prints `arm64` (Apple Silicon) or `x86_64` (Intel): the processor type the app will run on. If `PyInstaller` is missing, run `python -m pip install "pyinstaller>=6.0"`.

4. Allow the bundled engines and build the app (this takes a few minutes):

```
xattr -dr com.apple.quarantine bin/macos
chmod +x bin/macos/rnarobo bin/macos/knotty
python build_app.py
```

The output should end with `Ad-hoc codesign successful.`, `Signature verified OK.` and `DMG created: dist/RSAS_macOS.dmg`.

5. Test it:

```
open dist/RSAS.app
```

Check that the About page shows version 4.0.0, that **Load Example → Run** produces results, that the RNAfold Deck creates a PowerPoint file, and that the RNArobo Search and Pseudoknot Finder pages show a green banner. On Apple Silicon these two pages need Rosetta 2, because the bundled engines are Intel builds: `softwareupdate --install-rosetta --agree-to-license`. Then open `dist/RSAS_macOS.dmg` and double-click **Install RSAS.sh**, as a user would.

6. Distribute it. Rename the DMG to show the processor type, for example `RSAS_4.0.0_macOS_arm64.dmg` or `RSAS_4.0.0_macOS_intel.dmg`, and attach it to the GitHub release. To support both kinds of Mac, build once on each. Do not commit `dist/`; it is excluded by `.gitignore`.

> **Note**
>
> The app is signed ad hoc but not notarized by Apple, so users may see *"RSAS is damaged and can't be opened"*. **Install RSAS.sh** in the DMG removes the quarantine flag; the other solutions are in §7.4. Notarization, which removes the warning entirely, requires an Apple Developer account.

7.3 Linux

Step 1. Install Python with Tk and venv support.

Distribution	Command
Ubuntu / Debian	`sudo apt update && sudo apt install -y python3 python3-venv python3-pip python3-tk`
Fedora / RHEL / Rocky / Alma	`sudo dnf install -y python3 python3-pip python3-tkinter`
openSUSE	`sudo zypper install -y python3 python3-pip python3-tk`
Arch / Manjaro	`sudo pacman -S --needed python python-pip tk`

Check that the system Python is 3.10 or later (`python3 --version`). On older distributions (for example RHEL 8 or Ubuntu 20.04), install a newer Python, such as `python3.11` from the distribution's extra repositories, or use Method A.

Step 2. Create the environment (in the RSAS folder):

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

Step 3. Verify and run:

```
python -m RnaThermofinder version
python main.py
```

On headless servers, use the command line only; no display is required. To run the GUI on a server, use SSH X forwarding (`ssh -X`) or a virtual display (`xvfb-run python main.py`).

7.4 macOS security warnings (Gatekeeper)

RSAS is academic software and is not notarized by Apple. When you open the packaged **RSAS.app** (from the `.dmg` release) or when RSAS runs the bundled `rnarobo` and `knotty` executables in `bin/macos/`, macOS may refuse with one of these messages:

- *"RSAS" is damaged and can't be opened. You should move it to the Trash.*
- *"RSAS" can't be opened because Apple cannot check it for malicious software.*
- *"rnarobo" cannot be opened because the developer cannot be verified.*

The files are not damaged. macOS marks every file downloaded from the internet with a `com.apple.quarantine` attribute. When a quarantined program is opened, **Gatekeeper** checks whether it is signed and notarized by Apple; if it is not, macOS blocks it and shows the (misleading) "damaged" message. Use the solutions below in order, and only for copies of RSAS downloaded from the official repository.

Solution 1. Allow it in System Settings (recommended, safest).

1. Open RSAS once; when the error appears, click **Cancel** (or **Done**).
2. Open **System Settings → Privacy & Security**.
3. Scroll down to **Security**. A message says that "RSAS" (or "rnarobo" / "knotty") was blocked because it is not from an identified developer.
4. Click **Open Anyway** and enter your Mac password. RSAS opens, and macOS remembers the choice.

On macOS 15 (Sequoia) and later, right-click → **Open** no longer bypasses this check; use **Open Anyway** as above.

Solution 2. Remove the quarantine flag in Terminal (if **Open Anyway** does not appear):

```
# The application installed in /Applications
sudo xattr -cr /Applications/RSAS.app

# The RNArobo and Knotty executables in your RSAS folder (no sudo needed: it is your own
folder)
cd /path/to/RSAS
xattr -dr com.apple.quarantine bin/macos
chmod +x bin/macos/rnarobo bin/macos/knotty
```

Tip: type `sudo xattr -cr ` (with a trailing space) and drag **RSAS.app** from Finder into the Terminal window to fill in its path. Then open RSAS again. If the "damaged" message still appears, go to **System Settings → Privacy & Security**, where **Open Anyway** should now be available (Solution 1, steps 3–4).

The DMG also contains **Install RSAS.sh**, which copies RSAS.app to `/Applications` and removes the quarantine flag in one step.

If Terminal reports "Operation not permitted", it lacks permission to change the file:

1. Open **System Settings → Privacy & Security → Full Disk Access**.
2. Unlock if needed, click **+**, choose **Applications → Utilities → Terminal.app**, and turn its switch on.
3. Quit Terminal completely, open it again, and rerun the `xattr` command.

Solution 3. Temporarily disable Gatekeeper (if the commands above fail). Re-enable it immediately after RSAS has opened once:

```
sudo spctl --master-disable     # recent macOS versions also ask you to confirm in Privacy &
Security ("Anywhere")
open /Applications/RSAS.app
sudo spctl --master-enable      # turn Gatekeeper back on right away
```

Solution 4. Ad-hoc code signing (last resort). Sign the program locally with an ad-hoc signature:

```
sudo codesign --force --deep --sign - /Applications/RSAS.app
codesign --force --sign - bin/macos/rnarobo bin/macos/knotty      # from your RSAS folder
```

Step	Method	When to use it
1	System Settings → Privacy & Security → **Open Anyway**	First choice; safest
2	`xattr` (remove quarantine), then **Open Anyway**	**Open Anyway** is not offered at first; also the standard one-time step for `bin/macos/`
3	Disable Gatekeeper temporarily	The Terminal commands fail; re-enable immediately
4	Ad-hoc code signing	Last resort

Running RSAS from source (Conda or `python main.py`) does not trigger these warnings for RSAS itself, because Python is already trusted; only the `rnarobo` and `knotty` executables in `bin/macos/` need Solution 2 once.

---

8. Optional engines: RNArobo and Knotty

Only two pages depend on external executables: **RNArobo Search** (`rnarobo`) and **Pseudoknot Finder** (`knotty`). Everything else, including the main analysis and the RNAfold Deck, works without them.

> **Note**
>
> **No administrator rights are needed.** `bin/` below always means the `bin/` folder **inside your RSAS folder** (for example `~/RSAS/bin/linux/`), not a system folder such as `/usr/bin`. Every command in this section writes only to your RSAS folder or to a temporary folder.

RSAS looks for each executable in this order:

1. inside the packaged application (PyInstaller build);
2. `bin/<platform>/` in your RSAS folder: `bin/macos/`, `bin/linux/` or `bin/windows/`;
3. the `PATH` (a user-owned folder on the `PATH`, such as `~/.local/bin`, also works).

Knotty also needs its energy parameter folder, `simfold/`, next to the `knotty` executable. Each page shows a green banner with the path of the engine when it is found.

Engine	Authors and source	License	Reference
RNArobo 2.1.0	Computational Biology, Comenius University in Bratislava: https://compbio.fmph.uniba.sk/software.php (newest version: https://github.com/rampasek/RNArobo)	GPL-3.0	Rampášek L, Jimenez RM, Lupták A, Vinař T, Brejová B. *RNA motif search with data-driven element ordering.* BMC Bioinformatics 17:216 (2016)
Knotty	Hosna Jabbari, University of Alberta: https://github.com/HosnaJabbari/Knotty (energy parameters from SimFold/MultiRNAFold)	GPL	Jabbari H, Wark I, Montemagno C, Will S. *Knotty: efficient and accurate prediction of complex RNA pseudoknot structures.* Bioinformatics 34:3849–3856 (2018)

In the commands below, replace `/path/to/RSAS` with your RSAS folder (the folder that contains `main.py`).

8.1 macOS (bundled)

Both executables ship in `bin/macos/` together with Knotty's parameters. They are Intel (x86_64) builds:

```
cd /path/to/RSAS

# Apple Silicon only: install Rosetta 2 once
softwareupdate --install-rosetta --agree-to-license

# Remove the download quarantine flag so macOS lets the binaries run
xattr -dr com.apple.quarantine bin/macos
chmod +x bin/macos/rnarobo bin/macos/knotty
```

If macOS still blocks `rnarobo` or `knotty` (*cannot be opened because the developer cannot be verified*), allow them in **System Settings → Privacy & Security → Open Anyway**, or follow the other solutions in §7.4.

To use native Apple Silicon builds instead, build both engines from source (§8.4) and copy them into `bin/macos/`, replacing the bundled files.

8.2 Linux and WSL2

RNArobo: download the ready-made Linux executable (64-bit) published by its authors, straight into your RSAS folder:

```
cd /path/to/RSAS
mkdir -p bin/linux
curl -L -o bin/linux/rnarobo https://compbio.fmph.uniba.sk/rnarobo/rnarobo-2.1.0-linux64
#   (or: wget -O bin/linux/rnarobo https://compbio.fmph.uniba.sk/rnarobo/rnarobo-2.1.0-linux64)
chmod +x bin/linux/rnarobo
bin/linux/rnarobo 2>&1 | head -n 5          # prints its usage
```

If this executable does not run on your distribution, build RNArobo from source (§8.4.1).

Knotty: build it from source and install it in `bin/linux/` (§8.4.2). No ready-made Linux executable is published.

Docker. To use the engines inside the container, mount the folder that contains `rnarobo`, `knotty` and `simfold/`:

```
docker run --rm -it -v "$PWD/bin/linux":/opt/rsas/bin/linux ... rsas:4.0.0 gui
```

8.3 Windows

Both engines ship for Windows in `bin\windows\`: `rnarobo.exe`, `knotty.exe`, and Knotty's energy parameters in `bin\windows\simfold\`. RSAS finds them automatically; nothing has to be installed. Keep the `simfold\` folder next to `knotty.exe`.

If Windows SmartScreen or an antivirus program blocks `rnarobo.exe` or `knotty.exe`, choose **More info → Run anyway** or allow the file in Windows Security. As an alternative, run RSAS inside **WSL2** (Ubuntu) and follow §8.2.

8.4 Building from source (macOS and Linux)

You need a C/C++ compiler and `make`:

System	Command (once)
macOS	`xcode-select --install` (Apple Command Line Tools)
Debian / Ubuntu / WSL2	`sudo apt install -y build-essential`
Fedora / RHEL / Rocky	`sudo dnf install -y gcc gcc-c++ make`

On a shared server without administrator rights, the compilers are usually already installed (`gcc --version`), or available as a module (`module load gcc`).

Set two variables for the commands below (`PLAT` is `macos` or `linux`):

```
RSAS=/path/to/RSAS
PLAT=linux                        # or: PLAT=macos
mkdir -p "$RSAS/bin/$PLAT"
```

8.4.1 RNArobo from source

From the release archive (RNArobo 2.1.0):

```
cd /tmp
curl -LO https://compbio.fmph.uniba.sk/rnarobo/rnarobo-2.1.0.tar.gz
tar -zxf rnarobo-2.1.0.tar.gz
cd rnarobo-2.1.0
make
cp rnarobo "$RSAS/bin/$PLAT/"
chmod +x "$RSAS/bin/$PLAT/rnarobo"
```

or from the newest version on GitHub:

```
cd /tmp
git clone https://github.com/rampasek/RNArobo.git
cd RNArobo
make
cp rnarobo "$RSAS/bin/$PLAT/"
```

The RNArobo manual (descriptor format) is at https://compbio.fmph.uniba.sk/rnarobo/rnarobo-readme.pdf.

8.4.2 Knotty from source

Knotty needs a compiler that supports C++11 (g++ 4.7.2 or later, or Apple clang) and **CMake 3.1 or later**. Check with `cmake --version`. If CMake is missing or older, install it with any of:

```
conda install -n rsas -c conda-forge cmake      # inside the RSAS Conda environment (no
administrator rights)
python -m pip install --user cmake              # any Python, user install (no administrator
rights)
brew install cmake                              # macOS with Homebrew (https://brew.sh)
sudo apt install -y cmake                       # Debian / Ubuntu / WSL2
```

Download, build and install Knotty. Build it in a folder you keep (here `~/tools/Knotty-master`), because Knotty's authors note that its `simfold/` folder must not be deleted or moved after compiling:

```
mkdir -p ~/tools && cd ~/tools
curl -L -o Knotty-master.zip https://github.com/HosnaJabbari/Knotty/archive/master.zip
unzip -q Knotty-master.zip
cd Knotty-master                 # the folder that contains Knotty's README.md

cmake -H. -Bbuild
cmake --build build
#   C++11 errors? Delete the build folder and specify the compiler:
#   rm -rf build && cmake -H. -Bbuild -DCMAKE_CXX_COMPILER=g++ && cmake --build build

# Install into your RSAS folder: the executable and its energy parameters, side by side
cp "$(find . -maxdepth 2 -type f -name knotty -perm -u+x | head -n 1)" "$RSAS/bin/$PLAT/"
cp -R simfold "$RSAS/bin/$PLAT/"
chmod +x "$RSAS/bin/$PLAT/knotty"
```

`git clone https://github.com/HosnaJabbari/Knotty.git` can replace the download and `unzip` lines. If you recompile after moving the Knotty folder, delete its `build/` folder first.

8.5 Checking the engines

```
cd /path/to/RSAS
python -c "from RnaThermofinder.utils.rnarobo_wrapper import check_rnarobo_available as c;
print(c())"
python -c "from RnaThermofinder.utils.knotty_wrapper import check_knotty_available as c;
print(c())"
(cd bin/linux && ./knotty GCAACGAUGACAUACAUCGCUAGUCGACGC -w)     # bin/macos on a Mac
```

Each `check_…` line prints `(True, '<path of the executable>')`, and the Knotty test prints a dot-bracket structure with its energy. In the application, the RNArobo Search and Pseudoknot Finder pages show a green banner.

---

9. Verifying the installation

Run these checks inside the activated environment, from the repository root:

```
# 1. Versions
python -m RnaThermofinder version

# 2. Dependencies
python -c "import RNA, numpy, openpyxl, Bio, customtkinter, pptx, matplotlib;
print('dependencies OK')"

# 3. Test suite (includes a check that the local fold reproduces the RNAfold webserver exactly)
python -m pytest -q tests

# 4. End-to-end analysis of the bundled example
python -m RnaThermofinder analyze Examples/sample_dataset.fasta --out-dir ./rsas_check
```

Expected results:

Check	Expected
1	`RSAS 4.0.0 | ViennaRNA 2.7.x | Python 3.1x.x`
2	`dependencies OK`
3	`71 passed` (or more)
4	A folder `rsas_check/sample_dataset_<timestamp>/` with `sample_dataset_rna_results.csv`, `…_rna_results.xlsx`, `…_run_log.txt` and `…_run_info.json`

Finally, start the GUI with `python main.py`. The sidebar should list nine pages: Analyze, Results, Settings, Sequence Extractor, Synthetic Pool, RNArobo Search, Pseudoknot Finder, RNAfold Deck and About.

---

10. Running RSAS

Task	Command
Desktop application	`python main.py` (or `rsas`, or `python -m RnaThermofinder gui`)
Analysis without a display	`python -m RnaThermofinder analyze input.fasta [--out-dir DIR] [--cores N] [--temps 25,37,42] [--hairpin terminal|rbs_based]`
RNAfold Deck without a display	`python -m RnaThermofinder deck --results RUN/input_rna_results.xlsx --tier "Tier 1" --fold both --temp 37 --out-dir RUN --name input_RNAfold_deck`
Versions	`python -m RnaThermofinder version`

`rsas-cli` is equivalent to `python -m RnaThermofinder` after `pip install -e .`.

Output layout. Every analysis creates its own folder, and every file in it starts with the input file's name (without the extension). Earlier results are never overwritten.

```
~/RSAS/                                  # default; change with --out-dir
└── my_sequences_20261103_151813/
    ├── my_sequences_rna_results.csv
    ├── my_sequences_rna_results.xlsx    # tabs: Full Sequence, Hairpin Analysis (+ Motif
Matches)
    ├── my_sequences_run_log.txt
    ├── my_sequences_run_info.json       # input, versions, temperatures, settings
    └── my_sequences_RNAfold_deck_…/     # created by the RNAfold Deck page
```

RNArobo and Knotty exports go to `~/RSAS/<input>_rnarobo_<timestamp>/` and `~/RSAS/<input>_knotty_<timestamp>/`. User settings are stored in `~/.rsas/`.

On a cluster, the command line runs in batch jobs. For example, with SLURM:

```
#!/bin/bash
#SBATCH --job-name=rsas --cpus-per-task=8 --mem=8G --time=04:00:00
source "$HOME/miniforge3/etc/profile.d/conda.sh"
conda activate rsas
cd /path/to/RSAS
python -m RnaThermofinder analyze /path/to/input.fasta --out-dir /path/to/results --cores
"$SLURM_CPUS_PER_TASK"
```

---

11. Building a standalone application

A self-contained application, which needs no Python on the target machine, is built with PyInstaller on each target operating system (cross-compilation is not supported). Step-by-step instructions: Building the Windows executable (§7.1) and Building the macOS application (§7.2). In short:

```
python -m pip install -r requirements-dev.txt     # adds pytest, setuptools, pyinstaller
python build_app.py
```

Platform	Output
macOS	`dist/RSAS.app` and a distributable `.dmg` (RNArobo and Knotty included)
Windows	`dist/RSAS.exe`

Platform	Output
Linux	`dist/RSAS` (place Linux builds of the engines in `bin/linux/` first to include them)

The application is not notarized by Apple. On first launch, macOS may report that *RSAS is damaged and can't be opened*; the app is not damaged, and §7.4 explains how to open it (System Settings → Privacy & Security → **Open Anyway**, or removing the quarantine flag). Windows SmartScreen requires **More info → Run anyway**.

---

12. Updating and uninstalling

Update:

```
git pull                                    # or download the new release
conda env update -f environment.yml --prune      # Conda (environment-windows.yml on Windows)
python -m pip install -r requirements.txt --upgrade   # native venv
docker build -t rsas:<new-version> .              # Docker
```

Uninstall:

Method	Command
Conda	`conda deactivate && conda env remove -n rsas`
Native	Delete the `.venv` folder in the RSAS directory
Docker	`docker image rm rsas:4.0.0`

Then delete the RSAS folder. User settings in `~/.rsas/` and results in `~/RSAS/` are kept until you delete them yourself.

---

13. Troubleshooting

Symptom	Cause and solution
`ModuleNotFoundError: No module named 'RNA'`	ViennaRNA is not installed in the active environment. Activate it (`conda activate rsas` or `source .venv/bin/activate`) and run `python -m pip install ViennaRNA` (venv) or `conda install -c bioconda viennarna` (Conda, Linux/macOS).
`pip` cannot find a ViennaRNA version for your Python	ViennaRNA wheels exist for Python 3.10–3.13. Create the environment with Python 3.11 or 3.12.

Symptom	Cause and solution
`ModuleNotFoundError: No module named 'tkinter'` / `_tkinter`	Python was installed without Tk. Linux: install `python3-tk` (Debian/Ubuntu), `python3-tkinter` (Fedora) or `tk` (Arch). macOS: use the python.org installer or `brew install python-tk@3.x`. Windows: rerun the Python installer, choose *Modify*, and tick *tcl/tk and IDLE*.
`_tkinter.TclError: no display name and no $DISPLAY environment variable`	No graphical display (SSH session, server, container). Use the command line, `ssh -X`, or `xvfb-run python main.py`.
`conda env create` fails on Windows with `viennarna ... does not exist`	`environment.yml` was used on Windows. Use `environment-windows.yml`.
GUI widgets look broken or the theme is wrong	customtkinter 6.x was installed. Run `python -m pip install "customtkinter>=5.2,<6"`.
RNAfold Deck needs 'pptx' is shown on the RNAfold Deck page	Deck dependencies are missing. Run `python -m pip install python-pptx matplotlib pillow lxml`, or update the Conda environment.
RNArobo Search / Pseudoknot Finder shows a red "not found" banner	See §8. On macOS, run the `xattr` and Rosetta commands. On Windows, check that `bin\windows\` contains `rnarobo.exe`, `knotty.exe` and `simfold\`. On Linux, download RNArobo and build Knotty into `bin/linux/` of your RSAS folder (no administrator rights needed).
macOS: *"RSAS" is damaged and can't be opened*	The app is not damaged; macOS blocks programs that are not notarized by Apple. System Settings → Privacy & Security → **Open Anyway**, or `sudo xattr -cr /Applications/RSAS.app`. All options: §7.4.
macOS: *"rnarobo" cannot be opened because the developer cannot be verified*	Run `xattr -dr com.apple.quarantine bin/macos` once in your RSAS folder, or use **Open Anyway** (§7.4).
macOS: `xattr` reports *Operation not permitted*	Give Terminal **Full Disk Access** (System Settings → Privacy & Security), restart Terminal and run the command again (§7.4).
Knotty runs but returns an error about parameters	`simfold/params/` is missing next to the `knotty` executable. Copy `bin/macos/simfold` to the same folder as your `knotty`.
`PermissionError` when writing results	The output folder is not writable. Choose another folder with `--out-dir`, or make `~/RSAS` writable. In Docker, see the `--user` note in §6.3.
Docker on macOS: `cannot open display`	XQuartz is not running, or *Allow connections from network clients* is off. Rerun `xhost + 127.0.0.1` after restarting XQuartz.

Symptom	Cause and solution
Apple Silicon: a Conda package is not found for `osx-arm64`	Create an Intel environment instead: `CONDA_SUBDIR=osx-64 conda env create -f environment.yml`, then `conda activate rsas && conda config --env --set subdir osx-64`.

If a problem persists, open an issue and attach the output of `python -m RnaThermofinder version`, your operating system, the installation method, and the `<input>_run_log.txt` of the failing run.

---

14. Validation status of this guide

Procedure	Status
`environment.yml`: dependency resolution on linux-64, linux-aarch64, osx-64 and osx-arm64	✅ Verified (ViennaRNA 2.7.2, Python 3.11)
`environment-windows.yml`: dependency resolution on win-64; pip wheels available	✅ Verified
Conda environment on Linux: creation, 71 tests, CLI analysis, RNAfold Deck, GUI start	✅ Verified
Native venv + pip on Linux (Python 3.12): installation, 71 tests, GUI start, `pip install -e .` commands	✅ Verified
ViennaRNA PyPI wheels for Windows x64, macOS Intel/Apple Silicon, Linux x86_64/aarch64 (Python 3.10–3.13)	✅ Verified on PyPI
Native installation run on Windows and macOS hardware	⏳ To be confirmed by the first user on each platform
`docker build` and the Docker commands	⏳ The Dockerfile has not yet been built; base images could not be pulled in the validation environment
RNArobo / Knotty downloads and builds (§8.2, §8.4)	⏳ Commands taken from the authors' pages and READMEs; not yet run

When you confirm one of the pending procedures on your machine, please update this table.